Simultaneous quadratic-cost and bottleneck bounds for planar random matching

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Abstract

For two independent samples of N uniform points in the unit square, we construct a single matching whose total squared distance is $O(\log N)$ and whose longest edge is $O((\log N)^{3/4}/\sqrt{N})$, with failure probability $O((\log N)^{-2})$. The matching can be chosen measurably so that its expected quadratic cost is $O(\log N)$ and its bottleneck bound fails with probability at most $C\exp(-c(\log N)^{3/2})$. We also obtain these conclusions for a random sample matched to fixed evenly spread points.

The proof combines the classical bottleneck matching theorem with a Fourier estimate for smoothed empirical measures. Lipschitz truncation of a scalar stream function gives one divergence solution with both an L^2 bound and an L^∞ bound. A continuity equation with uniformly positive densities converts this field into a coupling that retains both estimates. Rounding preserves the distance constraint and produces a permutation with the required quadratic cost.

1 Statement of the main result

Let $X_1,\dots,X_N,Y_1,\dots,Y_N$ be independent and uniformly distributed on $[0,1]^2$, and write $d(x,y)=|x-y|$ for Euclidean distance. The set of permutations of $\{1,\dots,N\}$ is denoted by $\mathfrak{S}_N$. We prove the following statement.

Theorem 1.1. *There are universal constants $C_0,C_1,C_2>0$ such that, for all sufficiently large N, with probability at least $1-C_2/(\log N)^2$, there exists $\pi\in\mathfrak{S}_N$ satisfying*

$$\sum_{i=1}^{N} d(X_i,Y_{\pi(i)})^2 \le C_0\log N, \qquad \max_{1\le i\le N} d(X_i,Y_{\pi(i)}) \le \frac{C_1(\log N)^{3/4}}{\sqrt{N}}.$$

Remark 1.2. For the same permutation, Cauchy–Schwarz gives

$$\sum_{i=1}^{N} d(X_i,Y_{\pi(i)}) \le \sqrt{N\sum_{i=1}^{N} d(X_i,Y_{\pi(i)})^2} \le \sqrt{C_0}\sqrt{N\log N}.$$

Thus, with $L=\max\{\sqrt{C_0},C_1\}$,

$$\lim_{N\to\infty}\mathbb{P}\left(\exists\pi\in\mathfrak{S}_N:\ \begin{array}{c}\displaystyle\sum_{i=1}^{N} d(X_i,Y_{\pi(i)}) \le L\sqrt{N\log N},\\[2ex] \displaystyle\max_{1\le i\le N} d(X_i,Y_{\pi(i)}) \le \frac{L(\log N)^{3/4}}{\sqrt{N}}\end{array}\right)=1.$$

This is the simultaneous conclusion in [4, Problem 5.3]. The corresponding fixed-target formulation in [3, Question 4.9.1] is treated in Corollary 6.2, including its expected total-distance bound. The full ultimate matching conjecture [3, Research Problem 17.1.2] asks for stronger coordinatewise exponential integrability. Even its symmetric exponent-four case is not established by the quadratic-cost and maximal-distance estimates proved here.

The total-distance and bottleneck orders separately are classical; see [3, Theorems 4.5.1 and 4.7.1]. The unconstrained quadratic matching cost is also understood: Ambrosio, Stra, and Trevisan [1, Theorem 1.1] prove that its expectation is asymptotic to $(2\pi)^{-1}\log N$ for two uniform square samples. The point of Theorem 1.1 is to obtain the quadratic order while imposing the bottleneck constraint on the same permutation.

Ouimet's preprint [5, Theorem 2.1] gives simultaneous total-distance and bottleneck bounds. Its planar flow interpolation [5, Lemmas 2.9–2.10] uses a closely related stream-function truncation. The present quadratic estimate is stronger than that theorem's stated conclusion: multiplying its total-distance and maximal-distance bounds gives only

$$\sum_i d(X_i, Y_{\pi(i)})^2 \le \big(\max_i d(X_i, Y_{\pi(i)})\big) \sum_i d(X_i, Y_{\pi(i)}) = O((\log N)^{5/4}).$$

Here the conversion through smooth densities retains the L^2 energy all the way to the final matching. We state the deterministic compatibility result in Proposition 3.5. In §6, we also derive expected-cost bounds and the fixed-target version from the same estimates.

2 Reduction to the torus and proof overview

Let $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$ be the flat torus with geodesic distance $d_{\mathbb{T}^2}$. Its ordinary Lebesgue measure has total mass 4, so the uniform probability density is $1/4$. We first state the torus version of the matching theorem.

Theorem 2.1 (Matching on the torus)**.** *Let $X_1, \dots, X_N, Y_1, \dots, Y_N$ be independent uniform points on $\mathbb{T}^2$. There are universal constants $C_0, C_1, C_2 > 0$ such that, for all sufficiently large N, with probability at least $1 - C_2/(\log N)^2$, there exists $\pi \in \mathfrak{S}_N$ satisfying*

$$\sum_{i=1}^N d_{\mathbb{T}^2}(X_i, Y_{\pi(i)})^2 \le C_0 \log N, \qquad \max_{1\le i\le N} d_{\mathbb{T}^2}(X_i, Y_{\pi(i)}) \le \frac{C_1(\log N)^{3/4}}{\sqrt{N}}.$$

Proof of Theorem 1.1 assuming Theorem 2.1. Let $X_1, \dots, X_N, Y_1, \dots, Y_N$ have the torus law in Theorem 2.1. We fold these samples onto the square using the map $F : \mathbb{T}^2 \to [0,1]^2$ defined by

$$F(z_1, z_2) = \big(d_{\mathbb{R}/2\mathbb{Z}}(z_1, 0),\, d_{\mathbb{R}/2\mathbb{Z}}(z_2, 0)\big).$$

On the representatives $[-1,1)^2$, this is simply $F(z_1, z_2) = (|z_1|, |z_2|)$. A uniform torus point has independent coordinates uniform on $[-1,1)$, whose absolute values are independent and uniform on $[0,1]$. Consequently, $F(X_1), \dots, F(X_N), F(Y_1), \dots, F(Y_N)$ have exactly the joint distribution of the square samples in Theorem 1.1.

Distance to a fixed point is a 1-Lipschitz function on any metric space. Applying this fact to each circle coordinate gives

$$|F(z) - F(z')|^2 \le \sum_{\ell=1}^2 d_{\mathbb{R}/2\mathbb{Z}}(z_\ell, z'_\ell)^2 = d_{\mathbb{T}^2}(z, z')^2.$$

Thus every matching of the torus samples induces a matching of the folded samples with no increase in any edge length. On the event in Theorem 2.1, the same permutation π therefore satisfies

$$\sum_{i=1}^{N} |F(X_i) - F(Y_{\pi(i)})|^2 \le C_0 \log N, \qquad \max_{1\le i\le N} |F(X_i) - F(Y_{\pi(i)})| \le \frac{C_1 (\log N)^{3/4}}{\sqrt{N}}.$$

The event that the folded samples admit such a permutation contains the torus matching event, so its probability is at least $1 - C_2/(\log N)^2$. The equality of the joint sample laws proves Theorem 1.1 with the same constants. □

It remains to prove Theorem 2.1. We sketch its proof, emphasizing how the quadratic-cost and maximal-distance bounds are obtained for a single matching. Throughout the rest of the paper, $X_1, \dots, X_N, Y_1, \dots, Y_N$ denote the independent uniform torus samples in that theorem. We first construct a coupling of two smooth densities and then transfer its bounds to a permutation.

We smooth each empirical measure at radius

$$h = A\sqrt{\frac{\log N}{N}},$$

where A is a sufficiently large fixed constant. With probability tending to one, the resulting densities ρ_0, ρ_1 are bounded above and below by positive constants. Two separate constructions then give solutions of the equation $\operatorname{div} u = \rho_0 - \rho_1$. A Fourier calculation yields a field u with $\|u\|_2^2 = O(\log N/N)$. The bottleneck matching theorem, followed by a common smoothing of the matched endpoints, gives a coupling with distance cutoff $r = O((\log N)^{3/4}/\sqrt{N})$. From this coupling we construct another field v, with the same divergence, satisfying $\|v\|_\infty = O(r)$.

The main deterministic step combines these two fields. After subtracting their spatial means, the divergence-free difference $u - v$ is the rotated gradient of a scalar stream function. On the set where the maximal function of $|u|$ is controlled, this stream function is Lipschitz with constant $O(\|v\|_\infty)$. Extending its restriction from that set gives a field w with the same divergence and with

$$\|w\|_2^2 \le C\,\|u\|_2^2, \qquad \|w\|_\infty \le C\,\|v\|_\infty .$$

The area of the modified set controls the additional energy. This is the step that makes the two estimates compatible.

We next interpolate linearly between ρ_0 and ρ_1, using w as a fixed flux. The lower density bound controls both the speed and the quadratic action of the resulting flow. Its endpoint coupling therefore has quadratic cost $O(\log N/N)$ and distance cutoff $O(r)$. Passing from the smoothed measures back to the empirical measures adds at most $2h$ to the cutoff and $O(h^2)$ to the normalized quadratic cost. These errors have the required sizes because $h = o(r)$ and $h^2 = O(\log N/N)$. A decomposition into permutation matrices then selects one permutation with both bounds.

The deterministic inputs are collected in §3, culminating in Proposition 3.5. In §4 we establish the density and energy estimates and apply the Leighton–Shor bottleneck theorem [2], in the form stated in [3, Theorem 4.7.1]. The rounding argument in §5 then completes the proof of Theorem 2.1. The further consequences in §6 use a matching that minimizes quadratic cost subject to a fixed distance cutoff. The density, energy, and bottleneck estimates hold together with probability at least $1 - C/(\log N)^2$, which gives the stated failure probability. Constants denoted by $C, c > 0$ may change between occurrences. They are independent of N, and may depend on the fixed mollifier introduced below.

3 Preliminary inputs

This section develops the deterministic ingredients of the proof. The first lemma combines two solutions of the same divergence equation. The second constructs a bounded solution from a coupling with a distance cutoff. The third turns a field satisfying both bounds into a single coupling, leading to Proposition 3.5. The common setting and normalizations are fixed here for the remainder of the paper.

We use the torus and geodesic distance defined in §2. All spatial norms and integrals below are taken with respect to ordinary Lebesgue measure on $\mathbb{T}^2$, whose total mass is 4; the uniform probability density is therefore $1/4$. For an integrable scalar function or vector field g, its spatial mean is $\overline{g} = \frac{1}{4}\int_{\mathbb{T}^2} g$. All divergence identities are interpreted distributionally. For integrable f, the identity $\operatorname{div} u = f$ means

$$-\int_{\mathbb{T}^2} u \cdot \nabla\varphi \,\mathrm{d}x = \int_{\mathbb{T}^2} f\varphi \,\mathrm{d}x \qquad \text{for every smooth periodic test function } \varphi.$$

A coupling of two probability densities is a probability measure on $\mathbb{T}^2 \times \mathbb{T}^2$ with those densities as its marginals. We say it has distance cutoff r when its support is contained in $\{(x, y) : d_{\mathbb{T}^2}(x, y) \le r\}$.

Fix a nonnegative function $\eta \in C_c^\infty(\mathbb{R}^2)$, supported in the unit ball and with integral one. For $0 < s < 1/8$, write

$$\eta_s^{\mathbb{R}}(z) = s^{-2}\eta(z/s), \qquad \eta_s(z) = \sum_{m\in\mathbb{Z}^2} \eta_s^{\mathbb{R}}(z + 2m).$$

Thus η_s is a smooth probability kernel on $\mathbb{T}^2$, supported where $d_{\mathbb{T}^2}(z, 0) \le s$. Convolution on the torus uses ordinary Lebesgue measure, with $(\eta_s * g)(x) = \int_{\mathbb{T}^2} \eta_s(x - y)g(y)\,\mathrm{d}y$. For a finite measure the same notation denotes convolution against that measure. Integrals over $\mathbb{R}^2$ use $\eta_s^{\mathbb{R}}$, and periodic functions appearing in such integrals are evaluated periodically. No symmetry assumption on η is needed.

3.1 Combining two divergence solutions

The purpose of the next lemma is to retain an L^2 bound from one divergence solution and an L^∞ bound from another. We first recall the analytic estimates used in its proof. For an integrable function g, define the periodic Hardy–Littlewood maximal function by

$$\mathcal{M}g(x) = \sup_{s>0} \frac{1}{|B_{\mathbb{T}^2}(x, s)|} \int_{B_{\mathbb{T}^2}(x,s)} |g(y)|\,\mathrm{d}y.$$

Lebesgue differentiation gives $|g| \le \mathcal{M}g$ almost everywhere, and $\mathcal{M}g \le \|g\|_\infty$ for bounded g. The weak maximal inequality and $(\mathcal{M}g)^2 \le \mathcal{M}(|g|^2)$ imply

$$\left|\{x \in \mathbb{T}^2 : \mathcal{M}g(x) > t\}\right| \le \frac{C}{t^2}\,\|g\|_2^2, \qquad g \in L^2(\mathbb{T}^2), \quad t > 0.$$

We also use the pointwise Sobolev estimate

$$|\psi(x) - \psi(y)| \le C d_{\mathbb{T}^2}(x, y)(\mathcal{M}(|\nabla\psi|)(x) + \mathcal{M}(|\nabla\psi|)(y)), \qquad \psi \in H^1(\mathbb{T}^2), \tag{3.1}$$

for every pair of Lebesgue points x, y of the chosen Sobolev representative. In particular, a bound on the maximal function of the gradient makes the restriction of ψ Lipschitz.

The Euclidean version and its use in Lipschitz approximation are given in [7, proof of Lemma 10, p. 236]. The same argument applies on the fixed torus. To see this, let ψ_B denote the average of ψ over a ball B. Poincaré's inequality, followed by telescoping the averages over $B_{\mathbb{T}^2}(x, 2^{-j}s)$, gives

$$|\psi(x) - \psi_{B_{\mathbb{T}^2}(x,s)}| \le Cs\,\mathcal{M}(|\nabla\psi|)(x).$$

For $s = d_{\mathbb{T}^2}(x, y)$, compare the averages over $B_{\mathbb{T}^2}(x, s)$ and $B_{\mathbb{T}^2}(y, s)$ with the average over $B_{\mathbb{T}^2}(x, 3s)$. The volume-doubling property and Poincaré's inequality bound their difference by the right-hand side of (3.1). The preceding telescoping estimate at both endpoints then proves that inequality. Large radii are covered by the global Poincaré inequality on $\mathbb{T}^2$. The periodic weak maximal inequality follows from the usual covering argument on this compact doubling space. All constants depend only on the fixed torus.

Lemma 3.1 (Simultaneous bounds for a divergence solution)**.** *Suppose $u \in L^2(\mathbb{T}^2;\mathbb{R}^2)$ and $v \in L^\infty(\mathbb{T}^2;\mathbb{R}^2)$ have the same distributional divergence. There exists $w \in L^2(\mathbb{T}^2;\mathbb{R}^2) \cap L^\infty(\mathbb{T}^2;\mathbb{R}^2)$ such that*

$$\operatorname{div} w = \operatorname{div} u, \qquad \|w\|_2^2 \le C\,\|u\|_2^2\,, \qquad \|w\|_\infty \le C\,\|v\|_\infty\,.$$

Proof. We first remove the constant parts of the fields. Replacing u, v by $u - \overline{u}, v - \overline{v}$ preserves their divergences, and

$$\|u - \overline{u}\|_2^2 = \|u\|_2^2 - 4|\overline{u}|^2 \le \|u\|_2^2\,, \qquad \|v - \overline{v}\|_\infty \le 2\,\|v\|_\infty\,.$$

It is therefore enough to prove the result for mean-zero fields, which we continue to denote by u, v. This reduction removes the constant divergence-free fields that cannot be represented by a periodic stream function.

The field $u - v$ belongs to L^2, has zero mean, and is divergence free. Hence it is a rotated gradient:

$$u - v = \nabla^\perp\psi, \qquad \nabla^\perp\psi = (-\partial_2\psi, \partial_1\psi), \qquad \psi \in H^1(\mathbb{T}^2).$$

For completeness, use the Fourier convention

$$\widehat{g}(k) = \frac{1}{4}\int_{\mathbb{T}^2} g(z)e^{-i\pi k\cdot z}\,\mathrm{d}z, \qquad k \in \mathbb{Z}^2.$$

For every $k \neq 0$, the divergence condition says that $\widehat{(u - v)}(k)$ is orthogonal to k. With $k^\perp = (-k_2, k_1)$, define

$$\widehat{\psi}(k) = \frac{k^\perp \cdot \widehat{(u - v)}(k)}{i\pi|k|^2} \quad (k \neq 0), \qquad \widehat{\psi}(0) = 0.$$

The Fourier coefficient of $\nabla^\perp\psi$ is then exactly $\widehat{(u - v)}(k)$. Parseval's identity gives $\|\nabla\psi\|_2 = \|u - v\|_2$, and the absence of the zero mode also gives $\psi \in L^2$. Thus this formula defines the required real-valued, mean-zero function in $H^1(\mathbb{T}^2)$.

Write $B = \|v\|_\infty$. If $B = 0$, the common divergence is zero and we take $w = 0$. Suppose $B > 0$, and define

$$G = \{x \in \mathbb{T}^2 : \mathcal{M}(|u|)(x) \le B\},$$

intersected with the full-measure set where the representatives and (3.1) are valid. On G, the stream-function identity implies

$$\mathcal{M}(|\nabla\psi|) \le \mathcal{M}(|u|) + \mathcal{M}(|v|) \le 2B.$$

It follows from (3.1) that $\psi|_G$ is CB-Lipschitz with respect to $d_{\mathbb{T}^2}$.

If G is nonempty, extend this restriction by the McShane formula [7, Lemma 5]:

$$\widetilde{\psi}(x) = \inf_{y \in G} \{\psi(y) + CB\, d_{\mathbb{T}^2}(x, y)\}.$$

Here the constant is chosen to dominate the Lipschitz constant of the restriction. The infimum is finite: after fixing one point of G, the Lipschitz bound and the finite diameter of $\mathbb{T}^2$ bound ψ above and below on G. The same bound shows that $\widetilde{\psi} = \psi$ on G and that $\widetilde{\psi}$ is globally CB-Lipschitz on $\mathbb{T}^2$. In particular, it is periodic and $\left\|\nabla\widetilde{\psi}\right\|_\infty \le CB$. If G is empty, set $\widetilde{\psi} = 0$.

Define $w = v + \nabla^\perp\widetilde{\psi}$. Distributional derivatives commute, so $\operatorname{div}\nabla^\perp\widetilde{\psi} = 0$. Therefore $\operatorname{div} w = \operatorname{div} v = \operatorname{div} u$, and $\|w\|_\infty \le CB$. Moreover, locality of weak derivatives gives $\nabla\widetilde{\psi} = \nabla\psi$ almost everywhere on G, and consequently $w = u$ there. This locality property applies to a measurable agreement set. Indeed, the difference of the two Sobolev functions has an absolutely continuous representative on almost every line parallel to either coordinate axis. Its derivative vanishes almost everywhere on its zero set on each such line; Fubini's theorem gives the asserted equality of the weak gradients on G.

The weak maximal estimate bounds the area of the set where the field was changed:

$$|G^c| \le \frac{C}{B^2}\,\|u\|_2^2\,.$$

Combining this with $w = u$ on G and the global bound $\|w\|_\infty \le CB$, we obtain

$$\|w\|_2^2 = \int_G |u|^2 + \int_{G^c} |w|^2 \le \|u\|_2^2 + CB^2|G^c| \le C\,\|u\|_2^2\,.$$

This calculation also applies when G is empty. Finally, the initial mean removal changes the input bounds by at most the fixed factors already recorded. The resulting w already has the divergence of the original fields, so no constant field is added back. □

Remark 3.2. The exceptional set is determined by the field u, whose energy is small, while the truncation level is determined by $\|v\|_\infty$. The product of the squared truncation level and the area of this set is at most $C\,\|u\|_2^2$, which is why the modification preserves the energy bound. Modifying a scalar stream function preserves divergence exactly. No bound on $\|v\|_2$ in terms of $\|u\|_2$, or on $\|u\|_\infty$, is required.

This is an application of classical Lipschitz truncation. Schiffer's periodic truncation theorem for closed differential forms [6, Theorem 5.1] implies the same simultaneous bounds for divergence solutions on a torus of any fixed dimension. The proof above gives an elementary planar argument and makes the roles of the two input fields explicit.

3.2 From a distance cutoff to a bounded flux

We next turn a coupling with uniformly bounded displacement into a bounded solution of the divergence equation. The natural flux follows the shortest paths joining the coupled points. These paths can concentrate at intermediate times, so we first average their flux at the scale of the distance cutoff. The endpoint density bound controls the averaged field, and a local correction then restores its exact divergence.

Lemma 3.3 (A bounded divergence solution from a distance cutoff)**.** *Let ρ_0, ρ_1 be probability densities on $\mathbb{T}^2$ satisfying $0 \le \rho_0, \rho_1 \le M$ almost everywhere. Suppose a coupling γ of $\rho_0\,\mathrm{d}x$ and*

$\rho_1 \,\mathrm{d}x$ is supported on $\{(x,y) : d_{\mathbb{T}^2}(x,y) \le r\}$, where $0 < r < 1/8$. There exists $v \in L^\infty(\mathbb{T}^2;\mathbb{R}^2)$ such that

$$\operatorname{div} v = \rho_0 - \rho_1, \qquad \|v\|_\infty \le CMr.$$

The divergence identity holds in the sense of distributions.

Proof. Set $f = \rho_0 - \rho_1$. For γ-almost every (x,y), let $\delta(x,y) \in \mathbb{R}^2$ be the shortest displacement vector from x to y, so that $y = x + \delta(x,y)$ modulo $2\mathbb{Z}^2$. The restriction on r makes this vector unique, with $|\delta(x,y)| = d_{\mathbb{T}^2}(x,y) \le r$. All points on the path $x + t\delta(x,y)$, $0 \le t \le 1$, are interpreted on the torus.

Define the averaged path flux by

$$V(z) = \int_{\mathbb{T}^2\times\mathbb{T}^2} \delta(x,y) \int_0^1 \eta_r(z - x - t\delta(x,y)) \,\mathrm{d}t \,\mathrm{d}\gamma(x,y).$$

For fixed r, the kernel and all its derivatives are bounded, so V is smooth and differentiation under the integral is valid. The identity

$$\delta(x,y) \cdot \nabla_z \eta_r(z - x - t\delta(x,y)) = -\frac{\mathrm{d}}{\mathrm{d}t} \eta_r(z - x - t\delta(x,y))$$

therefore gives

$$\begin{aligned} \operatorname{div} V(z) &= \int_{\mathbb{T}^2\times\mathbb{T}^2} (\eta_r(z-x) - \eta_r(z-y)) \,\mathrm{d}\gamma(x,y) \\ &= (\eta_r * \rho_0)(z) - (\eta_r * \rho_1)(z) = (\eta_r * f)(z). \end{aligned}$$

Here the second equality uses the two marginals of γ.

To bound V, fix z. If a kernel in its defining integral is nonzero, then $d_{\mathbb{T}^2}(z, x + t\delta(x,y)) \le r$, and the triangle inequality implies $d_{\mathbb{T}^2}(z,x) \le 2r$. Since $\|\eta_r\|_\infty \le Cr^{-2}$, we obtain

$$\begin{aligned} |V(z)| &\le Cr^{-2} r \,\gamma\big(B_{\mathbb{T}^2}(z,2r) \times \mathbb{T}^2\big) \\ &= Cr^{-1} \int_{B_{\mathbb{T}^2}(z,2r)} \rho_0(x) \,\mathrm{d}x \le CMr. \end{aligned}$$

In the last step we used the area bound $|B_{\mathbb{T}^2}(z,2r)| \le Cr^2$. Thus the estimate only involves the mass of starting endpoints near z; no density bound on the intermediate path locations is needed.

It remains to correct the difference between f and its convolution. Define the periodic vector field

$$A_f(z) = \int_{\mathbb{R}^2} \eta_r^{\mathbb{R}}(a)\, a \int_0^1 f(z - ta) \,\mathrm{d}t \,\mathrm{d}a.$$

This expression defines an essentially bounded field because $f \in L^\infty$ and $\eta_r^{\mathbb{R}}$ is supported on $|a| \le r$. We verify its divergence directly, so that no derivative of f is required. For a smooth periodic test function φ, translation invariance of Lebesgue measure and Fubini's theorem give

$$\begin{aligned} -\int_{\mathbb{T}^2} A_f(z) \cdot \nabla\varphi(z) \,\mathrm{d}z &= -\int_{\mathbb{R}^2} \eta_r^{\mathbb{R}}(a) \int_{\mathbb{T}^2} f(q) \int_0^1 a \cdot \nabla\varphi(q + ta) \,\mathrm{d}t \,\mathrm{d}q \,\mathrm{d}a \\ &= \int_{\mathbb{R}^2} \eta_r^{\mathbb{R}}(a) \int_{\mathbb{T}^2} f(q)(\varphi(q) - \varphi(q+a)) \,\mathrm{d}q \,\mathrm{d}a \\ &= \int_{\mathbb{T}^2} (f(z) - (\eta_r * f)(z))\varphi(z) \,\mathrm{d}z. \end{aligned}$$

For the last equality, translate $z = q + a$ in the term containing $\varphi(q + a)$ and use the definition of periodization. Consequently, $\operatorname{div} A_f = f - \eta_r * f$ in distributions. This identity uses only the unit integral of the mollifier and is valid without any symmetry assumption on it.

Finally, $\|f\|_\infty \le M$, and hence

$$\|A_f\|_\infty \le \|f\|_\infty \int_{\mathbb{R}^2} \eta_r^{\mathbb{R}}(a)\,|a|\,\mathrm{d}a \le Mr.$$

The field $v = V + A_f$ has divergence f and satisfies $\|v\|_\infty \le CMr$, as claimed. □

3.3 A single coupling from both flux bounds

The final deterministic input converts a vector field with controlled energy and supremum norm into one coupling with the corresponding cost and distance bounds. A positive lower bound on the two endpoint densities permits a linear interpolation between them. Dividing the fixed flux by this interpolated density gives a velocity field whose speed and quadratic action are both controlled. This is the Dacorogna–Moser interpolation used for quadratic transport, as in [1, Proposition 2.3]; we record its simultaneous distance-cutoff conclusion and the approximation argument needed here.

Lemma 3.4 (A coupling with controlled distance and quadratic cost)**.** *Let ρ_0, ρ_1 be probability densities on $\mathbb{T}^2$ satisfying $\rho_0, \rho_1 \ge a > 0$ almost everywhere. Suppose $w \in L^2(\mathbb{T}^2;\mathbb{R}^2) \cap L^\infty(\mathbb{T}^2;\mathbb{R}^2)$ satisfies $\operatorname{div} w = \rho_0 - \rho_1$ in distributions. Then there is a coupling Γ of $\rho_0\,\mathrm{d}x$ and $\rho_1\,\mathrm{d}x$ such that*

$$\operatorname{supp}\Gamma \subseteq \{(x,y) : d_{\mathbb{T}^2}(x,y) \le a^{-1}\|w\|_\infty\}, \qquad \int d_{\mathbb{T}^2}(x,y)^2\,\mathrm{d}\Gamma(x,y) \le a^{-1}\|w\|_2^2.$$

Proof. First suppose that the densities and the field are smooth. For $0 \le t \le 1$, set

$$\rho_t = (1-t)\rho_0 + t\rho_1, \qquad b_t = \frac{w}{\rho_t}.$$

The interpolation satisfies $\rho_t \ge a$ and has total mass one at every time. Moreover, $\rho_t b_t = w$, so the divergence assumption is exactly the continuity equation

$$\partial_t \rho_t + \operatorname{div}(\rho_t b_t) = (\rho_1 - \rho_0) + \operatorname{div} w = 0.$$

The field b_t is smooth on the compact torus, and hence its flow F_t exists for $0 \le t \le 1$ and consists of diffeomorphisms, with $F_0(x) = x$ and $\partial_t F_t(x) = b_t(F_t(x))$.

We check that the flow carries the prescribed densities. Its Jacobian satisfies $\partial_t \det DF_t(x) = (\operatorname{div} b_t)(F_t(x)) \det DF_t(x)$. Thus the continuity equation implies

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\rho_t(F_t(x)) \det DF_t(x)\big) = \big(\partial_t\rho_t + \operatorname{div}(\rho_t b_t)\big)(F_t(x)) \det DF_t(x) = 0.$$

Evaluating at $t = 0$ gives $\rho_t(F_t(x)) \det DF_t(x) = \rho_0(x)$. Change of variables therefore yields $(F_t)_\#(\rho_0\,\mathrm{d}x) = \rho_t\,\mathrm{d}x$. In particular,

$$\Gamma = (\mathrm{id}, F_1)_\#(\rho_0\,\mathrm{d}x)$$

is a coupling of the required endpoint measures.

The geodesic distance between the endpoints of a trajectory is at most its length. Since $\|b_t\|_\infty \le a^{-1}\|w\|_\infty$, every initial point satisfies

$$d_{\mathbb{T}^2}(x, F_1(x)) \le \int_0^1 |b_t(F_t(x))|\,\mathrm{d}t \le a^{-1}\|w\|_\infty.$$

This gives the support constraint for Γ. For the same trajectory, Cauchy–Schwarz on the unit time interval gives $d_{\mathbb{T}^2}(x, F_1(x))^2 \le \int_0^1 |b_t(F_t(x))|^2 \,\mathrm{d}t$. Integrating this inequality against the initial density and using the pushforward identity at each time yields

$$\begin{aligned}\int d_{\mathbb{T}^2}(x,y)^2 \,\mathrm{d}\Gamma(x,y) &\le \int_0^1 \int_{\mathbb{T}^2} |b_t(F_t(x))|^2 \rho_0(x) \,\mathrm{d}x \,\mathrm{d}t \\ &= \int_0^1 \int_{\mathbb{T}^2} |b_t(y)|^2 \rho_t(y) \,\mathrm{d}y \,\mathrm{d}t \\ &= \int_0^1 \int_{\mathbb{T}^2} \frac{|w(y)|^2}{\rho_t(y)} \,\mathrm{d}y \,\mathrm{d}t \le a^{-1} \|w\|_2^2 .\end{aligned}$$

The support and cost estimates thus hold for the same flow coupling.

For general data, let $\rho_i^\varepsilon = \eta_\varepsilon * \rho_i$, $i = 0, 1$, and $w^\varepsilon = \eta_\varepsilon * w$. Convolving with the same nonnegative probability kernel preserves total mass, the lower bound $\rho_i^\varepsilon \ge a$, and the identity $\operatorname{div} w^\varepsilon = \rho_0^\varepsilon - \rho_1^\varepsilon$. Young's inequality also gives $\|w^\varepsilon\|_2 \le \|w\|_2$ and $\|w^\varepsilon\|_\infty \le \|w\|_\infty$. The smooth argument therefore produces couplings Γ_ε whose support and quadratic cost satisfy the stated bounds with the original norms of w, uniformly in ε.

Because $\mathbb{T}^2 \times \mathbb{T}^2$ is compact, a subsequence of Γ_ε converges weakly to a probability measure Γ as $\varepsilon \downarrow 0$. The approximate-identity property gives $\rho_i^\varepsilon \to \rho_i$ in $L^1(\mathbb{T}^2)$; testing against continuous functions of either coordinate identifies the marginals of Γ as $\rho_0 \,\mathrm{d}x$ and $\rho_1 \,\mathrm{d}x$. The common constraint set $\{(x,y) : d_{\mathbb{T}^2}(x,y) \le a^{-1} \|w\|_\infty\}$ is closed, so the limit is supported on this set. Finally, $d_{\mathbb{T}^2}(x,y)^2$ is bounded and continuous, and hence its integral converges along the subsequence. The quadratic-cost bound passes to Γ as well. □

Combining the three lemmas gives the deterministic statement used in the proof of the main theorem. The upper density bound converts the initial distance cutoff into a bounded flux, while the lower density bound controls the speed and action of the final interpolation.

Proposition 3.5 (Combining quadratic energy and bounded displacement)**.** *Let ρ_0, ρ_1 be probability densities on $\mathbb{T}^2$ with $a \le \rho_0, \rho_1 \le M$ almost everywhere, where $a > 0$. Suppose a coupling of these densities is supported on $\{(x,y) : d_{\mathbb{T}^2}(x,y) \le r\}$ for some $0 < r < 1/8$, and suppose $u \in L^2(\mathbb{T}^2; \mathbb{R}^2)$ satisfies $\operatorname{div} u = \rho_0 - \rho_1$ in distributions. There is a coupling Γ such that*

$$\operatorname{supp}\Gamma \subseteq \{(x,y) : d_{\mathbb{T}^2}(x,y) \le C(M/a)r\}, \qquad \int d_{\mathbb{T}^2}(x,y)^2 \,\mathrm{d}\Gamma(x,y) \le \frac{C}{a} \|u\|_2^2 .$$

Proof. Lemma 3.3 supplies a field v with $\operatorname{div} v = \rho_0 - \rho_1$ and $\|v\|_\infty \le CMr$. Apply Lemma 3.1 to u and v to obtain a field w satisfying

$$\operatorname{div} w = \rho_0 - \rho_1, \qquad \|w\|_2^2 \le C \|u\|_2^2 , \qquad \|w\|_\infty \le CMr.$$

Lemma 3.4, applied to this single field, gives a coupling whose distance cutoff is at most $C(M/a)r$ and whose quadratic cost is at most $Ca^{-1} \|u\|_2^2$. □

4 Regularizing independent uniform samples

We now verify the hypotheses of Proposition 3.5 for smoothed empirical measures of the torus samples in Theorem 2.1. We use $\mathbb{P}, \mathbb{E}$ for their joint law and expectation. With the mollifier fixed in §3, set

$$\mu_X = \frac{1}{N} \sum_{i=1}^N \delta_{X_i}, \qquad \mu_Y = \frac{1}{N} \sum_{j=1}^N \delta_{Y_j}, \qquad \rho_0 = \eta_h * \mu_X, \qquad \rho_1 = \eta_h * \mu_Y, \qquad h = A\sqrt{\frac{\log N}{N}}.$$

We choose A once and for all as in Lemma 4.1, and take N sufficiently large that $h < 1/8$. The three estimates below supply, respectively, the positive density bounds, a divergence solution with small energy, and a coupling with a distance cutoff.

4.1 Uniform bounds on the densities

Lemma 4.1. *For a sufficiently large fixed A,*

$$\mathbb{P}\left(\frac{1}{8} \le \rho_0, \rho_1 \le \frac{3}{8} \text{ everywhere on } \mathbb{T}^2\right) \ge 1 - CN^{-2}$$

for all sufficiently large N.

Proof. For a fixed z, the summands $\eta_h(z - X_i)$ are independent, with

$$\mathbb{E}\eta_h(z - X_i) = \frac{1}{4}, \qquad 0 \le \eta_h(z - X_i) \le Ch^{-2}, \qquad \operatorname{Var}(\eta_h(z - X_i)) \le Ch^{-2}.$$

For example, the variance bound follows from $\mathbb{E}\eta_h(z - X_i)^2 = \frac{1}{4}\int_{\mathbb{T}^2} \eta_h^2 \le Ch^{-2}$. Bernstein's inequality gives

$$\mathbb{P}\left(\left|\rho_0(z) - \frac{1}{4}\right| > \frac{1}{16}\right) \le 2\exp(-cNh^2).$$

Deterministically, $\|\nabla\rho_0\|_\infty \le Ch^{-3}$. Choose a torus net of mesh c_1h^3, with $c_1 > 0$ sufficiently small. It has at most Ch^{-6} points, and the density changes by at most $1/16$ between an arbitrary point and its nearest net point. A union bound therefore gives

$$\mathbb{P}\left(\sup_{z\in\mathbb{T}^2}\left|\rho_0(z) - \frac{1}{4}\right| > \frac{1}{8}\right) \le Ch^{-6}\exp(-cNh^2).$$

The same argument applies to ρ_1. Since $Nh^2 = A^2\log N$ and $h^{-6} \le C_A N^3$, choosing A large enough makes the combined failure probability at most CN^{-2}. □

4.2 A quadratic-energy estimate

Lemma 4.2. *Let $f = \rho_0 - \rho_1$, and define*

$$u = -\nabla(-\Delta)^{-1}f$$

using the inverse Laplacian on mean-zero periodic functions. Then $\overline{u} = 0$, $\operatorname{div} u = f$, and, for a fixed sufficiently large C,

$$\mathbb{P}\left(\|u\|_2^2 > C\frac{\log N}{N}\right) \le \frac{C}{(\log N)^2}.$$

Proof. The inverse Laplacian is well defined because both densities have integral one, so f has zero mean. We estimate both the mean and the variance of the energy. With the Fourier convention fixed in §3.1, Parseval's identity reads $\|g\|_2^2 = 4\sum_{k\in\mathbb{Z}^2}|\widehat{g}(k)|^2$. For the fixed function η on $\mathbb{R}^2$, write $\widehat{\eta}(\xi) = \int_{\mathbb{R}^2}\eta(z)e^{-i\xi\cdot z}\,\mathrm{d}z$. The Fourier coefficient of its periodic smoothing kernel is $\widehat{\eta_h}(k) = \frac{1}{4}\widehat{\eta}(\pi hk)$.

Pool the $2N$ points as $Z_1, \dots, Z_{2N}$, and put $\varepsilon_a = 1$ for $a \le N$ and $\varepsilon_a = -1$ for $a > N$. Then

$$\widehat{f}(k) = \frac{\widehat{\eta}(\pi hk)}{4N}\sum_{a=1}^{2N}\varepsilon_a e^{-i\pi k\cdot Z_a}, \qquad \widehat{f}(0) = 0.$$

The eigenvalue of $-\Delta$ at nonzero frequency k is $\pi^2|k|^2$. Parseval's identity therefore gives

$$E_h := \|u\|_2^2 = \frac{1}{N^2} \sum_{a,b=1}^{2N} \varepsilon_a \varepsilon_b K_h(Z_a - Z_b), \tag{4.1}$$

where the real, even, mean-zero kernel K_h is

$$K_h(z) = \frac{1}{4\pi^2} \sum_{k \in \mathbb{Z}^2 \setminus \{0\}} \frac{|\widehat{\eta}(\pi h k)|^2}{|k|^2} e^{i\pi k \cdot z}.$$

The coefficients of K_h are unchanged by $k \mapsto -k$, since η is real and $\widehat{\eta}(-\xi) = \overline{\widehat{\eta}(\xi)}$. Thus K_h is real and even without requiring η itself to be symmetric. Since $|\widehat{\eta}| \le 1$ and $\widehat{\eta}$ decays faster than any power, the two-dimensional lattice sum gives

$$K_h(0) \le C(1 + \log(1/h)).$$

Indeed, summing $|k|^{-2}$ over $1 \le |k| \le h^{-1}$ gives $O(1 + \log(1/h))$. Beyond h^{-1}, the Fourier decay bounds the remaining sum by a constant.

The diagonal part of (4.1) is the deterministic quantity $2K_h(0)/N$. All off-diagonal terms have expectation zero, because $Z_a - Z_b$ is uniform on $\mathbb{T}^2$ and K_h has zero mean. Hence

$$\mathbb{E}E_h = \frac{2K_h(0)}{N} \le C\frac{\log N}{N}.$$

An expectation estimate alone would not yield the desired conclusion with a fixed constant. To control its fluctuations, let Z, Z' be independent uniform torus points. Parseval gives

$$\mathbb{E}K_h(Z - Z')^2 = \frac{1}{16\pi^4} \sum_{k \ne 0} \frac{|\widehat{\eta}(\pi h k)|^4}{|k|^4} \le C.$$

Distinct unordered pairs $\{a, b\}$ and $\{c, d\}$ give uncorrelated off-diagonal terms. They are independent if they have no common index. If, for example, they share $a = c$ and their other indices differ, conditioning on Z_a shows that both conditional means are zero and the remaining variables are independent. Since K_h is even, we obtain exactly

$$\mathrm{Var}(E_h) = \frac{4}{N^4} \binom{2N}{2} \mathbb{E}K_h(Z - Z')^2 \le \frac{C}{N^2}.$$

Choose the threshold constant strictly larger than the constant in the expectation bound. Chebyshev's inequality then gives

$$\mathbb{P}\left(E_h > C\frac{\log N}{N}\right) \le \frac{C/N^2}{(\log N/N)^2} \le \frac{C}{(\log N)^2},$$

as required. □

4.3 The bottleneck input and common smoothing

Lemma 4.3 (Bottleneck matching input)**.** *There are universal constants $b, c, C > 0$ such that independent uniform samples on $\mathbb{T}^2$ satisfy*

$$\mathbb{P}\left(\exists \sigma \in \mathfrak{S}_N : \max_i d_{\mathbb{T}^2}(X_i, Y_{\sigma(i)}) \le b\,\frac{(\log N)^{3/4}}{\sqrt{N}}\right) \ge 1 - C\exp(-c(\log N)^{3/2}).$$

Proof. We first work in the unit square. The Leighton–Shor theorem [2], as stated in [3, Theorem 4.7.1], matches an independent uniform sample to any fixed evenly spread family of N points with radius at most $C(\log N)^{3/4}/\sqrt{N}$ and failure probability at most $C\exp(-c(\log N)^{3/2})$.

We specify a reference family valid for every N. In the definition of [3, §4.3], the square is partitioned into N rectangles of area $1/N$, each containing its prescribed point and lying within distance $10/\sqrt{N}$ of that point. To construct such a partition, let $m = \lfloor\sqrt{N}\rfloor$ and choose integers $n_1, \dots, n_m$, differing by at most one, whose sum is N. Make row j have height n_j/N, and divide it into n_j rectangles of width $1/n_j$. Each rectangle has area $1/N$. Since $m \le n_j \le m+2$, its width is at most $2/\sqrt{N}$ and its height is at most $3/\sqrt{N}$. Its diameter is therefore at most $\sqrt{13}/\sqrt{N}$, so the rectangle centers meet the cited definition.

Apply the theorem to each sample using these same centers. If τ_X, τ_Y map the two samples to the reference family, then $\sigma = \tau_Y^{-1} \circ \tau_X$ matches the samples to each other. The triangle inequality increases the radius by at most a factor two, and a union bound adds the two failure probabilities.

Finally, represent the uniform torus points in $[-1,1)^2$ and map this square to $[0,1)^2$ by scaling and translation. The preceding argument gives a matching whose Euclidean edge lengths in the original representatives are bounded by twice the unit-square bound. Torus distance is at most the Euclidean distance between representatives. Absorbing these fixed factors into b, c, C proves the lemma. □

For the remainder of the proof, let

$$r = b\,\frac{(\log N)^{3/4}}{\sqrt{N}}.$$

For all sufficiently large N, we have $r < 1/8$. If σ is the matching in Lemma 4.3, define a coupling of the smoothed densities by using the same shift at its two endpoints:

$$\gamma = \frac{1}{N}\sum_{i=1}^{N}\int_{\mathbb{R}^2} \eta_h^{\mathbb{R}}(z)\,\delta_{(X_i+z,\,Y_{\sigma(i)}+z)}\,\mathrm{d}z. \tag{4.2}$$

All points are understood modulo $2\mathbb{Z}^2$. The marginals are $\rho_0\,\mathrm{d}x, \rho_1\,\mathrm{d}x$. Translation invariance gives $d_{\mathbb{T}^2}(X_i+z, Y_{\sigma(i)}+z) = d_{\mathbb{T}^2}(X_i, Y_{\sigma(i)})$, so γ has the same cutoff r.

5 Returning to a permutation

To complete the proof of Theorem 2.1, we pass from a coupling of the smoothed densities to a matching of the original torus sample points. The conditional weights below assign each smoothed location to its empirical source. This gives a fractional matching supported on short edges. The Birkhoff–von Neumann theorem then selects a permutation while preserving that support and controlling the quadratic cost by averaging.

Lemma 5.1 (Rounding with a distance cutoff)**.** *Let $X_i, Y_j \in \mathbb{T}^2$, and let $\rho_0 = \eta_h * \mu_X$ and $\rho_1 = \eta_h * \mu_Y$ be positive. Suppose a coupling Γ of these densities satisfies*

$$\operatorname{supp}\Gamma \subseteq \{d_{\mathbb{T}^2}(x,y) \le D\}, \qquad \int d_{\mathbb{T}^2}(x,y)^2\,\mathrm{d}\Gamma(x,y) \le E.$$

There is a permutation $\pi \in \mathfrak{S}_N$ such that

$$\max_i d_{\mathbb{T}^2}(X_i, Y_{\pi(i)}) \le 2h + D, \qquad \frac{1}{N}\sum_i d_{\mathbb{T}^2}(X_i, Y_{\pi(i)})^2 \le 3(2h^2 + E).$$

Proof. The conditional weights

$$\alpha_i(x) = \frac{\eta_h(x - X_i)}{N\rho_0(x)}, \qquad \beta_j(y) = \frac{\eta_h(y - Y_j)}{N\rho_1(y)}$$

are nonnegative and satisfy $\sum_i \alpha_i(x) = \sum_j \beta_j(y) = 1$. Define the matrix

$$p_{ij} = \int_{\mathbb{T}^2 \times \mathbb{T}^2} \alpha_i(x)\beta_j(y)\, \mathrm{d}\Gamma(x, y).$$

Its row and column sums are $1/N$. For example,

$$\sum_j p_{ij} = \int_{\mathbb{T}^2} \alpha_i(x)\rho_0(x)\, \mathrm{d}x = \frac{1}{N}\int_{\mathbb{T}^2} \eta_h(x - X_i)\, \mathrm{d}x = \frac{1}{N}.$$

Wherever $\alpha_i(x)\beta_j(y) > 0$, the smoothing kernel implies $d_{\mathbb{T}^2}(X_i, x) \le h$ and $d_{\mathbb{T}^2}(Y_j, y) \le h$. Thus

$$p_{ij} > 0 \quad \Longrightarrow \quad d_{\mathbb{T}^2}(X_i, Y_j) \le 2h + D.$$

The squared triangle inequality, integrated against the same conditional weights, gives

$$\begin{aligned}\sum_{i,j} p_{ij} d_{\mathbb{T}^2}(X_i, Y_j)^2 &\le 3\int \sum_{i,j} \alpha_i(x)\beta_j(y)\big(d_{\mathbb{T}^2}(X_i, x)^2 + d_{\mathbb{T}^2}(x, y)^2 + d_{\mathbb{T}^2}(y, Y_j)^2\big)\, \mathrm{d}\Gamma \\ &\le 3(2h^2 + E).\end{aligned}$$

The matrix Np is doubly stochastic. By the Birkhoff–von Neumann theorem [8], it is a convex combination of permutation matrices. Every permutation with positive weight uses only edges for which $p_{ij} > 0$: a zero matrix entry cannot be a positive convex combination of nonnegative entries. Therefore all permutations in the decomposition satisfy the stated cutoff. At least one of them has normalized quadratic cost at most $\sum_{i,j} p_{ij} d_{\mathbb{T}^2}(X_i, Y_j)^2$, because that sum is the weighted average of their normalized costs. □

Proof of Theorem 2.1. Intersect the events of Lemmas 4.1, 4.2, and 4.3. Their intersection has probability at least

$$1 - CN^{-2} - \frac{C}{(\log N)^2} - C\exp(-c(\log N)^{3/2}) \ge 1 - \frac{C}{(\log N)^2}$$

for all sufficiently large N. No independence between these three events is required.

On this event, ρ_0, ρ_1 lie between $a = 1/8$ and $M = 3/8$, the field u has energy at most $C \log N/N$, and (4.2) is a coupling with cutoff r. Proposition 3.5 gives a single coupling Γ such that

$$\operatorname{supp}\Gamma \subseteq \{d_{\mathbb{T}^2}(x, y) \le Cr\}, \qquad \int d_{\mathbb{T}^2}(x, y)^2\, \mathrm{d}\Gamma(x, y) \le C\frac{\log N}{N}.$$

Apply Lemma 5.1. Since

$$h^2 = A^2\frac{\log N}{N}, \qquad \frac{h}{r} = \frac{A}{b(\log N)^{1/4}} \longrightarrow 0,$$

it gives one permutation satisfying

$$\sum_i d_{\mathbb{T}^2}(X_i, Y_{\pi(i)})^2 \le C\log N, \qquad \max_i d_{\mathbb{T}^2}(X_i, Y_{\pi(i)}) \le Cr.$$

Since $r = b(\log N)^{3/4}/\sqrt{N}$, choosing fixed C_0, C_1, C_2 large enough in the cost, cutoff, and failure-probability bounds proves Theorem 2.1. □

6 Expected cost and evenly spread targets

The preceding argument also controls the expected cost of a single matching while retaining the stronger probability estimate supplied by the bottleneck theorem. The same method applies when one sample is replaced by a fixed evenly spread family. In both cases, a finite optimization rule makes the choice of permutation measurable.

6.1 Expected cost and the bottleneck probability

Corollary 6.1. *In the square model of Theorem 1.1, there are universal constants $C, c > 0$ and a measurable choice of $\pi \in \mathfrak{S}_N$ such that, for all sufficiently large N,*

$$\mathbb{E} \sum_{i=1}^{N} d(X_i, Y_{\pi(i)})^2 \le C \log N,$$

and

$$\mathbb{P}\left(\max_i d(X_i, Y_{\pi(i)}) > C \frac{(\log N)^{3/4}}{\sqrt{N}} \right) \le C \exp\bigl(-c(\log N)^{3/2}\bigr).$$

The same choice of π satisfies both bounds in Theorem 1.1 with probability at least $1 - C/(\log N)^2$. In particular,

$$\mathbb{E} \sum_{i=1}^{N} d(X_i, Y_{\pi(i)}) \le C\sqrt{N \log N}.$$

Proof. Recall $r = b(\log N)^{3/4}/\sqrt{N}$, and fix a sufficiently large universal constant L. For each pair of square samples, choose a permutation minimizing the total squared cost among those whose longest edge is at most Lr. Break ties by a fixed lexicographic order, and choose the identity permutation if this constraint is infeasible. There are only finitely many permutations, and the feasibility and cost of each are continuous inequalities in the sample coordinates. Consequently, this rule is measurable. We choose L below, uniformly in N.

To analyze the rule, let $Z_1, \dots, Z_N, Z'_1, \dots, Z'_N$ be independent uniform torus points. The arrays $(F(Z_i))_{i=1}^N$ and $(F(Z'_i))_{i=1}^N$ have the joint law of the square samples. Let ρ_0, ρ_1 and E_h denote their torus smoothing densities and Poisson energy, as in §4. Let $\mathcal{B}$ be the torus bottleneck event from Lemma 4.3, and set

$$\mathcal{D} = \left\{ \frac{1}{8} \le \rho_0, \rho_1 \le \frac{3}{8} \right\}.$$

The existing estimates give

$$\mathbb{P}(\mathcal{D}^c) \le CN^{-2}, \qquad \mathbb{P}(\mathcal{B}^c) \le C \exp\bigl(-c(\log N)^{3/2}\bigr), \qquad \mathbb{E}E_h \le C \frac{\log N}{N}.$$

On $\mathcal{B} \cap \mathcal{D}$, the proof of Theorem 2.1, without imposing the energy event, constructs a torus permutation with cutoff $Cr+2h$ and total squared cost at most $CN(E_h+h^2)$. Indeed, Proposition 3.5 bounds the coupling cost by CE_h, and Lemma 5.1 then gives this estimate. Folding does not increase any edge length. Since $h = o(r)$, a fixed sufficiently large L makes this square permutation feasible for the selection rule, whose cost is therefore at most $CN(E_h + h^2)$.

On $\mathcal{B} \cap \mathcal{D}^c$, fold the original bottleneck permutation instead. It has cutoff r and total squared cost at most Nr^2, so the selected permutation has no greater cost. Thus the cutoff constraint is

feasible on all of $\mathcal{B}$. On $\mathcal{B}^c$, the selected permutation still has total squared cost at most $2N$, since the squared diameter of the unit square is 2. Taking expectations gives

$$\begin{aligned}\mathbb{E}\sum_i d(X_i, Y_{\pi(i)})^2 &\le CN(\mathbb{E}E_h + h^2) + Nr^2\mathbb{P}(\mathcal{D}^c) + 2N\mathbb{P}(\mathcal{B}^c)\\ &\le C\log N.\end{aligned}$$

Here $Nh^2 = O(\log N)$, $Nr^2 = O((\log N)^{3/2})$, and $N\exp(-c(\log N)^{3/2}) \to 0$.

Feasibility on $\mathcal{B}$ proves the claimed bottleneck probability. On $\mathcal{B}\cap\mathcal{D}\cap\{E_h \le C\log N/N\}$, the same selected permutation also has total squared cost at most $C\log N$. Lemmas 4.1, 4.2, and 4.3 bound the probability of its complement by $C/(\log N)^2$. Finally, Cauchy–Schwarz, first for the sum and then for expectation, gives

$$\mathbb{E}\sum_i d(X_i, Y_{\pi(i)}) \le \sqrt{N\,\mathbb{E}\sum_i d(X_i, Y_{\pi(i)})^2} \le C\sqrt{N\log N}.$$

□

6.2 Matching to fixed evenly spread points

We call a fixed family $Y_1, \dots, Y_N \in [0,1]^2$ *evenly spread* if there is a measurable partition $Q_1, \dots, Q_N$ of the square such that

$$|Q_j| = \frac{1}{N}, \qquad \sup_{y\in Q_j}|y - Y_j| \le \frac{10}{\sqrt{N}}, \qquad 1 \le j \le N.$$

This includes the rectangular partitions in the definition used in [3, §4.3]. Only the displayed area and distance conditions are needed below.

Corollary 6.2. *Let $Y_1, \dots, Y_N$ be a fixed evenly spread family, and let $X_1, \dots, X_N$ be independent uniform points in the unit square. There are universal constants $C, c > 0$, independent of the fixed family, and a measurable choice $\pi = \pi(X_1, \dots, X_N)$ such that, for all sufficiently large N,*

$$\mathbb{P}\left(\sum_i d(X_i, Y_{\pi(i)})^2 \le C\log N, \quad \max_i d(X_i, Y_{\pi(i)}) \le C\frac{(\log N)^{3/4}}{\sqrt{N}}\right) \ge 1 - \frac{C}{(\log N)^2}.$$

The same choice satisfies

$$\mathbb{E}\sum_i d(X_i, Y_{\pi(i)})^2 \le C\log N, \qquad \mathbb{E}\sum_i d(X_i, Y_{\pi(i)}) \le C\sqrt{N\log N},$$

and

$$\mathbb{P}\left(\max_i d(X_i, Y_{\pi(i)}) > C\frac{(\log N)^{3/4}}{\sqrt{N}}\right) \le C\exp\bigl(-c(\log N)^{3/2}\bigr).$$

Proof. Let $Z_1, \dots, Z_N$ be independent uniform torus points, so that $(F(Z_i))_{i=1}^N$ has the law of the square sample. We first couple the smoothed torus sample to uniform torus measure. Set

$$\rho_0 = \eta_h * \left(\frac{1}{N}\sum_i \delta_{Z_i}\right), \qquad \rho_1 = \frac{1}{4}, \qquad u = -\nabla(-\Delta)^{-1}(\rho_0 - \tfrac{1}{4}), \qquad E_h = \|u\|_2^2.$$

The proof of Lemma 4.1 gives

$$\mathbb{P}(\mathcal{D}^c) \le CN^{-2}, \qquad \mathcal{D} = \left\{ \frac{1}{8} \le \rho_0 \le \frac{3}{8} \right\}.$$

The Fourier calculation in Lemma 4.2, with N unsigned points in place of $2N$ signed points, gives

$$E_h = \frac{1}{N^2} \sum_{a,b=1}^{N} K_h(Z_a - Z_b).$$

The kernel K_h is the same real, even, mean-zero kernel as before. Its diagonal contribution is deterministic, and distinct unordered off-diagonal pairs are uncorrelated. Consequently,

$$\mathbb{E}E_h = \frac{K_h(0)}{N} \le C\frac{\log N}{N}, \qquad \operatorname{Var}(E_h) = \frac{4}{N^4}\binom{N}{2}\mathbb{E}K_h(Z - Z')^2 \le \frac{C}{N^2},$$

where Z, Z' are independent uniform torus points. Chebyshev's inequality therefore gives

$$\mathbb{P}\left(E_h > C\frac{\log N}{N} \right) \le \frac{C}{(\log N)^2}$$

for a sufficiently large fixed C.

Write $s = 10/\sqrt{N}$. The rectangular construction in the proof of Lemma 4.3, scaled to the torus, gives a fixed partition $\widetilde{Q}_1, \dots, \widetilde{Q}_N$ into cells of ordinary area $4/N$, with centers q_j, such that $d_{\mathbb{T}^2}(y, q_j) \le Cs$ for $y \in \widetilde{Q}_j$. Apply the one-sample Leighton–Shor theorem [3, Theorem 4.7.1] to these centers as in that proof. After increasing the universal constant b in the definition of r, if necessary, this gives an event $\mathcal{B}$ with

$$\mathbb{P}(\mathcal{B}^c) \le C\exp(-c(\log N)^{3/2})$$

on which some $\sigma \in \mathfrak{S}_N$ satisfies $\max_i d_{\mathbb{T}^2}(Z_i, q_{\sigma(i)}) \le r$. On this event, the measure

$$\Lambda = \frac{1}{N}\sum_i \delta_{Z_i} \otimes \left(\frac{N}{4}\mathbf{1}_{\widetilde{Q}_{\sigma(i)}}(y)\,\mathrm{d}y \right)$$

couples the torus empirical measure to $\mathrm{d}y/4$ and has cutoff $r + Cs$. Average its common translates $(x, y) \mapsto (x + a, y + a)$ against $\eta_h^{\mathbb{R}}(a)\,\mathrm{d}a$. The first marginal becomes $\rho_0\,\mathrm{d}x$, the second remains $\mathrm{d}y/4$, and translation invariance preserves the cutoff $r + Cs$.

Since $s = o(r)$ and $r + Cs < 1/8$ for sufficiently large N, Proposition 3.5 applies on $\mathcal{B} \cap \mathcal{D}$. It supplies a coupling Γ of $\rho_0\,\mathrm{d}x$ and $\mathrm{d}y/4$ such that

$$\operatorname{supp}\Gamma \subseteq \{d_{\mathbb{T}^2}(x, y) \le Cr\}, \qquad \int d_{\mathbb{T}^2}(x, y)^2\,\mathrm{d}\Gamma(x, y) \le CE_h.$$

To round this coupling to the prescribed square targets, use their partition $Q_1, \dots, Q_N$ and define

$$\alpha_i(x) = \frac{\eta_h(x - Z_i)}{N\rho_0(x)}, \qquad p_{ij} = \int \alpha_i(x)\mathbf{1}_{Q_j}(F(y))\,\mathrm{d}\Gamma(x, y).$$

The row sums are $1/N$, as in Lemma 5.1. The column sums are also $1/N$, because

$$\sum_i p_{ij} = \frac{1}{4}\int_{\mathbb{T}^2} \mathbf{1}_{Q_j}(F(y))\,\mathrm{d}y = |Q_j| = \frac{1}{N}.$$

Here we used that folding sends uniform torus measure to uniform square measure. On the support of a contributing weight,

$$|F(Z_i) - Y_j| \le d_{\mathbb{T}^2}(Z_i, x) + d_{\mathbb{T}^2}(x, y) + s \le h + Cr + s.$$

The same squared triangle inequality as before gives

$$\sum_{i,j} p_{ij} |F(Z_i) - Y_j|^2 \le 3(h^2 + CE_h + s^2).$$

The Birkhoff argument in Lemma 5.1 therefore produces a square permutation with cutoff $h + Cr + s$ and total squared cost at most $CN(E_h + h^2 + s^2)$.

We also need a fallback that does not require $\mathcal{D}$. On all of $\mathcal{B}$, round Λ directly by using instead the matrix

$$p_{ij} = \frac{1}{4} \int_{\widetilde{Q}_{\sigma(i)}} \mathbf{1}_{Q_j}(F(y))\, \mathrm{d}y.$$

Its row and column sums are again $1/N$. If $p_{ij} > 0$, then

$$|F(Z_i) - Y_j| \le r + Cs + s \le Cr.$$

Thus Birkhoff rounding gives a permutation with cutoff Cr and total squared cost at most CNr^2, regardless of the smoothed density bounds.

Choose the square permutation by minimizing total squared cost under the cutoff Lr, for a fixed universal L large enough to include both constructions above. Break ties lexicographically and use the identity permutation when the constraint is infeasible. This is the finite measurable selection rule used in Corollary 6.1; in particular, it depends only on the square sample and the fixed targets, not on a choice of torus lift. The folded sample law allows us to analyze it using the torus construction.

On $\mathcal{B} \cap \mathcal{D}$, optimality bounds its cost by $CN(E_h + h^2 + s^2)$. On $\mathcal{B} \cap \mathcal{D}^c$, the fallback bounds its cost by CNr^2; on $\mathcal{B}^c$, its cost is at most $2N$. It follows that

$$\begin{aligned} \mathbb{E} \sum_i d(X_i, Y_{\pi(i)})^2 &\le CN(\mathbb{E}E_h + h^2 + s^2) + CNr^2 \mathbb{P}(\mathcal{D}^c) + 2N\mathbb{P}(\mathcal{B}^c) \\ &\le C \log N. \end{aligned}$$

The expected total-distance bound follows by Cauchy–Schwarz. The cutoff constraint is feasible on $\mathcal{B}$, which gives the stated bottleneck probability. Finally, on $\mathcal{B} \cap \mathcal{D} \cap \{E_h \le C \log N/N\}$, the same permutation has total squared cost at most $C \log N$. The probability of the complement is at most $C/(\log N)^2$, proving the remaining assertion. All constants used the fixed target family only through the bound $s = 10/\sqrt{N}$, so they are uniform over all such families. □

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